Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Deep Learning, Anomaly Detection, Production ML | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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19th October 2026

Re: Applied Machine Learning Scientist II – TD

Dear Hiring Manager,

I am applying for the Applied Machine Learning Scientist II role in Analytics, Insights, & AI at TD. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems that run in production: data ingestion, modeling, deployment and monitoring. TD's work on turning complex data into enterprise analytic solutions, with business owners and engineers in the same room, is the way I have worked throughout my career.

Three parts of your posting match my work directly. For production ML, my published system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline at F1 0.76 (Journal of the Audio Engineering Society, 2026), and I owned its full lifecycle from data acquisition to deployment and monitoring. For anomaly detection on structured data, at Forestpin I designed and deployed ML and statistical pipelines for financial forensics and compliance monitoring, with SQL and Python scoring services, and turned compliance stakeholders' requirements into technical specs. For delivery at scale, at MAS Holdings I led computer vision systems for manufacturing that cut inspection time by 99.5% and raised operational efficiency by up to 300%, and I mentored engineers and set up code review and CI/CD.

I should be direct about the gaps. I have not used PySpark in production; my data work has been SQL, Python, Snowflake and Airflow pipelines on AWS, and I expect to pick up Spark quickly. My LLM experience is applied (RAG and tooling) rather than research-level, and I have not worked inside a bank, so I would learn TD's frameworks and risk and control practices from day one. I am comfortable explaining models to non-technical audiences, which I did with compliance and operations stakeholders.

I am a Canadian permanent resident based in Toronto, eligible to work without sponsorship, available immediately, and happy to work onsite in Toronto. I would welcome the opportunity to discuss how my experience can support the team at TD. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD